

Advances in Document Processing

A Comprehensive Survey

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Abstract

This report provides a comprehensive survey of document processing techniques, covering optical character recognition, layout analysis, semantic classification, and content extraction. We examine the evolution from rule-based systems to modern deep learning approaches, identifying key challenges in handling diverse document formats. Our analysis covers 150 papers published between 2015 and 2025, revealing trends toward end-to-end learning systems and multimodal architectures.

Chapter 1

Introduction

Document processing is the task of extracting structured information from unstructured document images or digital files. This field encompasses several sub-tasks including text recognition, layout analysis, and semantic understanding.

1.1 Motivation

The volume of digital documents continues to grow exponentially. Organizations need automated tools to extract, classify, and index information from PDF files, scanned documents, and other formats.

1.2 Scope

This survey focuses on digital PDF processing, excluding scanned document OCR. We examine techniques for preserving content fidelity during conversion to structured formats such as Markdown, HTML, and XML.

Chapter 2

Background

2.1 PDF Format

The Portable Document Format stores content as positioned drawing commands rather than logical text flow. This design prioritizes pixel-perfect rendering over content accessibility.

2.2 Markdown Format

Markdown is a lightweight markup language designed for readability. It supports headings, lists, tables, images, and inline formatting through simple syntax.